

Theorem Suite for Cosine–Riemann Ξ Mixtures

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For

$$H_t(z) = (1 - t) \cos z + t \Xi(z), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right), \quad 0 < t < 1,$$

this note records the current theorem-level COS– Ξ results. The 11 September localization manuscript remains the proof document for the first theorem and qualitative background, but it predates the later completeness, endpoint-front, and strip-transfer results.

Theorem C1: individual localization

Status: independent audit PASS.

Define

$$a = 2\pi e^3, \quad c_t = \log\left(2e^{1/2}\sqrt{\pi}\frac{t}{1-t}\right), \quad D_t = 3\log a + 2c_t - 3,$$
$$q_n = (2n + 1)\pi, \quad B(q) = W_0\left(-\frac{2iq}{a}\right), \quad u(q) = -\frac{2iq}{B(q)}, \quad h_t(q) = -3 - \frac{D_t}{B(q) + 1},$$

and

$$\widehat{z}_n(t) = i\left(u(q_n) + h_t(q_n) - \frac{1}{2}\right).$$

For each fixed $t \in (0, 1)$ there is $N(t)$ such that

$$|z - \widehat{z}_n(t)| < \frac{40}{(2n + 1)\pi}, \quad n \geq N(t),$$

contains exactly one zero $z_n(t)$ of H_t , and that zero is simple. The disks are eventually pairwise disjoint and ordered by real part. The threshold is uniform for t on compact subsets of $(0, 1)$.

Moreover

$$x_n(t) = \frac{4\pi n}{\log n} \left(1 + O_t\left(\frac{\log \log n}{\log n}\right)\right),$$
$$y_n(t) = \frac{2\pi^2 n}{(\log n)^2} \left(1 + O_t\left(\frac{\log \log n}{\log n}\right)\right),$$

and

$$y_n(t) = \frac{\pi x_n(t)}{2 \log x_n(t)} + O_t\left(\frac{x_n(t)}{(\log x_n(t))^2}\right).$$

Theorem C2: completeness and total nonreal count

Status: independent audit PASS.

For fixed t , the nonreal zero set is a finite exceptional multiset together with the fourfold symmetry orbits of the localized sequence:

$$\{\text{nonreal zeros of } H_t\} = E_t \uplus \biguplus_{n \geq N_C(t)} \{z_n, \bar{z}_n, -z_n, -\bar{z}_n\}.$$

Hence, with analytic multiplicity,

$$N_t^{\text{nr}}(R) \sim \frac{R \log R}{\pi}.$$

Theorem C3: fixed-height endpoint front

Status: independent audit PASS.

For fixed $d > 1/2$ put

$$\lambda = \log \frac{t}{1-t}, \quad R_d(t) = \inf\{|z| : H_t(z) = 0, \quad |\text{Im } z| \geq d\}.$$

Then

$$R_d(t) = \frac{4}{\pi} \lambda + \frac{2d+7}{\pi} \log \lambda + O_d(1)$$

as $\lambda \rightarrow +\infty$. The upper bound is realized by a simple first-quadrant root of height $d + O_d(1/\log \lambda)$.

Theorem C4: moving-boundary endpoint front

Status: S01 PASS_AS_WRITTEN.

Let

$$\beta(v) = \{54.004(\log v)^{2/3}(\log \log v)^{1/3}\}^{-1}.$$

There are absolute constants C, λ_0 such that for every $\lambda \geq \lambda_0$ and

$$1 - \frac{\beta(6\lambda)}{16} \leq s \leq \frac{3}{2}, \quad d = s - \frac{1}{2},$$

one has

$$\left| R_d(t) - \left\{ \frac{4}{\pi} \lambda + \frac{2d+7}{\pi} \log \lambda \right\} \right| \leq C(1 + \log \log(6\lambda)).$$

The same constants work across the whole moving interval. For the maximal allowed downward movement, the constructed mixture zero eventually satisfies $0 < \text{Im } z < 1/2$.

Theorem C5: fixed-strip cluster/count/phase transfer

Status: main theorem S02-T PASS_AS_WRITTEN.

Fix $0 < \delta < 1/2$, define

$$a = \frac{1}{2} + \delta, \quad h = \delta/4, \quad q = 1 - \delta/16, \quad r = T^{-2},$$

and the window

$$\Omega = \{\sigma - ix : a \leq \sigma \leq 2, \ T \leq x \leq 2T\}.$$

Let

$$J_q(Y) = 10395.2Y^q \log Y + 1.104(\log Y)^2 + 0.173 \log Y \log \log Y + 0.51 \log Y,$$

$$Q_\delta(T) = J_q(T/2) + J_q(T) + J_q(2T).$$

Under the explicit large-height hypotheses of the frozen proof and $\lambda \geq \Lambda(T)$:

1. total relevant zeta-zero multiplicity is at most $Q_\delta(T)$;
2. each full relevant component contains equal total mixture and ξ multiplicity M_j ;
- 3.

$$|N_H(\Omega) - N_\xi(\Omega)| \leq B_\Omega \leq Q_\delta(T), \quad N_H(\Omega) \leq Q_\delta(T) = o(T);$$

4. on zero-free paths, lifted phase changes differ by at most $2/(3T)$; on closed contours their windings agree exactly.

Audit exclusion. One redundant component-hole sentence in the original S02 proof was judged false as a general geometric statement. It is excluded here. The audit accepted the main S02-T transfer theorem without that inference.

Status and scope

C1, C2 and C3 are independently audited theorem blocks. C4 is S01 PASS_AS_WRITTEN. C5 is the accepted S02-T theorem. No theorem here is a proof of RH.